Design and Deployment of an Imaging Spectrometer for the Characterisation of Spatial Chirp

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Laser Plasma Accelerator Group
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Contents

Introduction

Plasma Modulated Plasma Accelerator

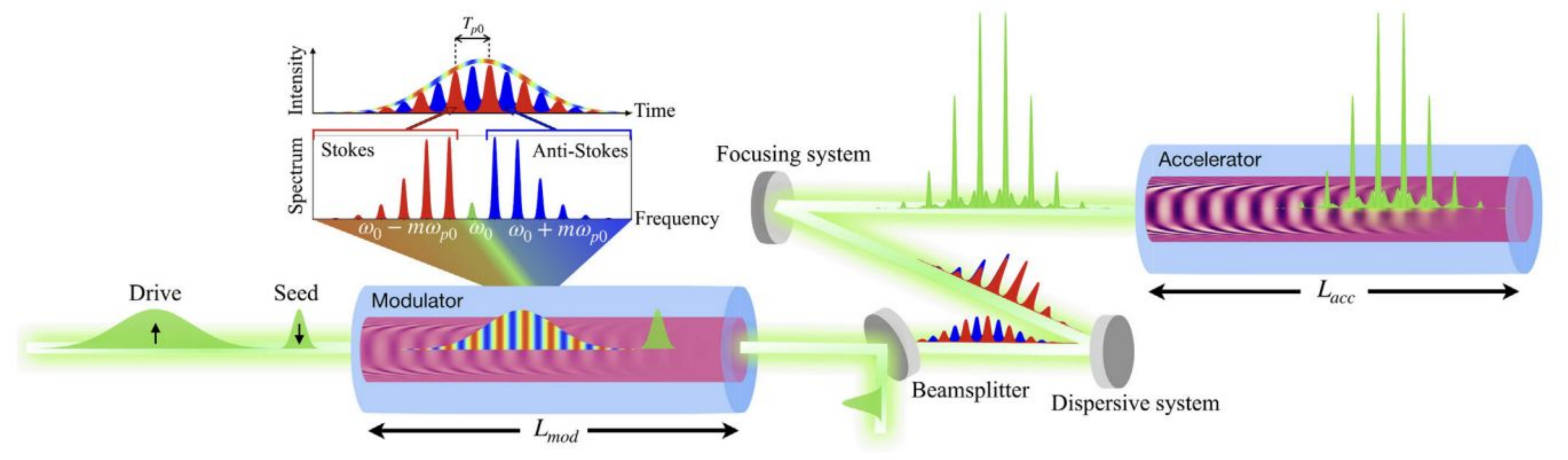


Figure sourced from [1]

Chirped Pulse Amplification

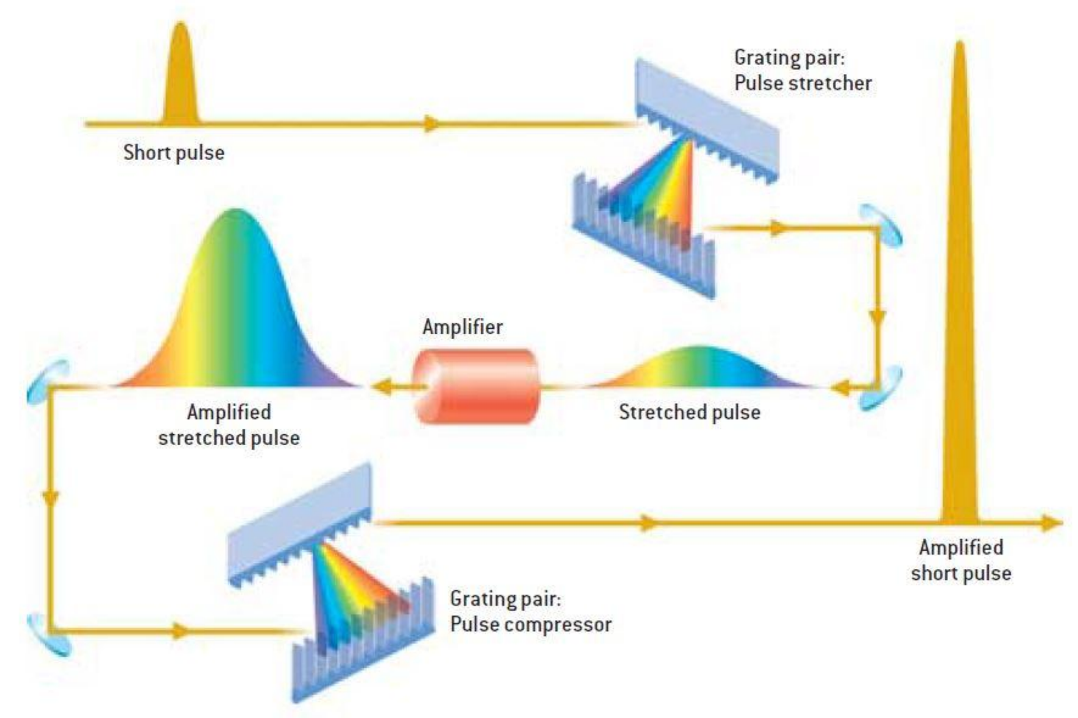
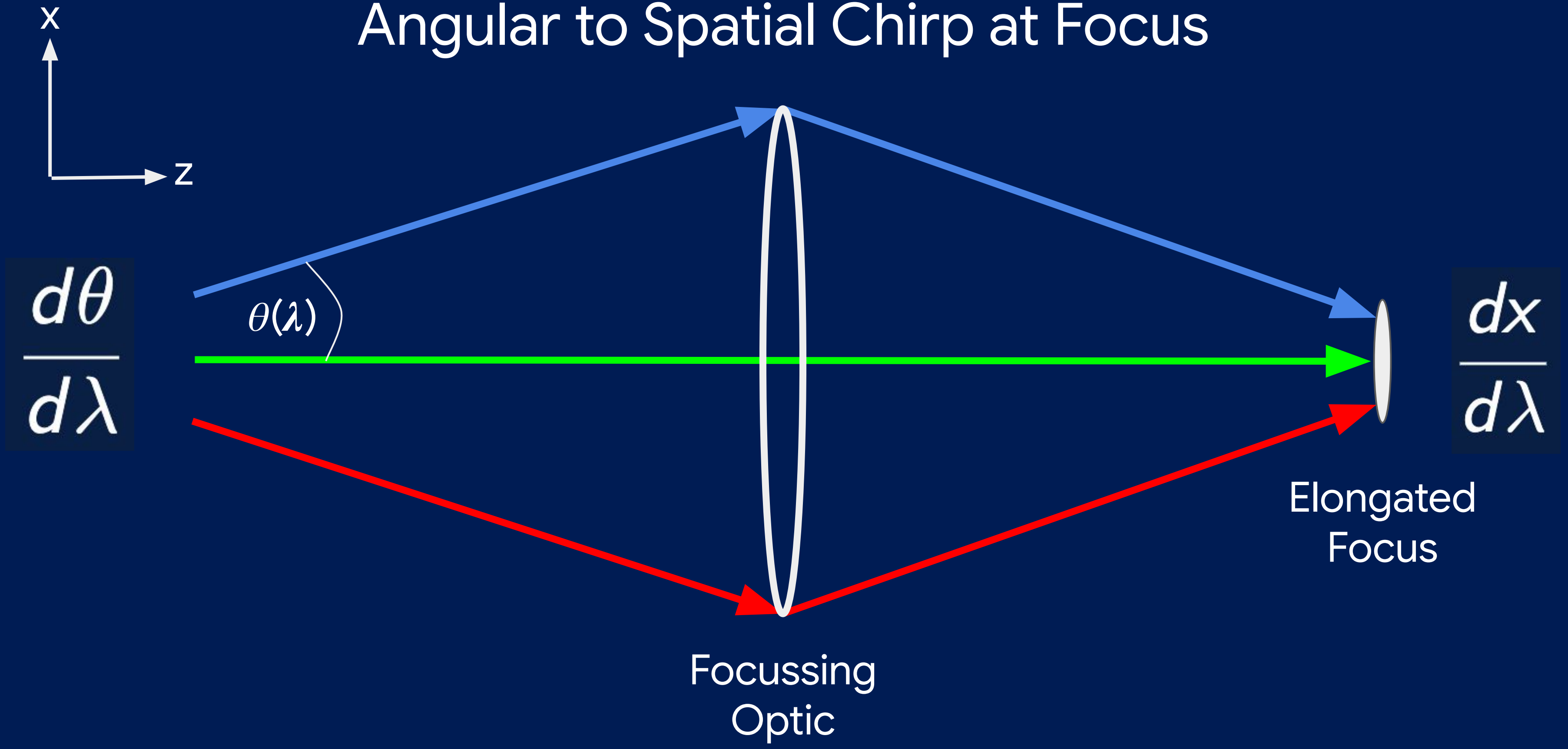


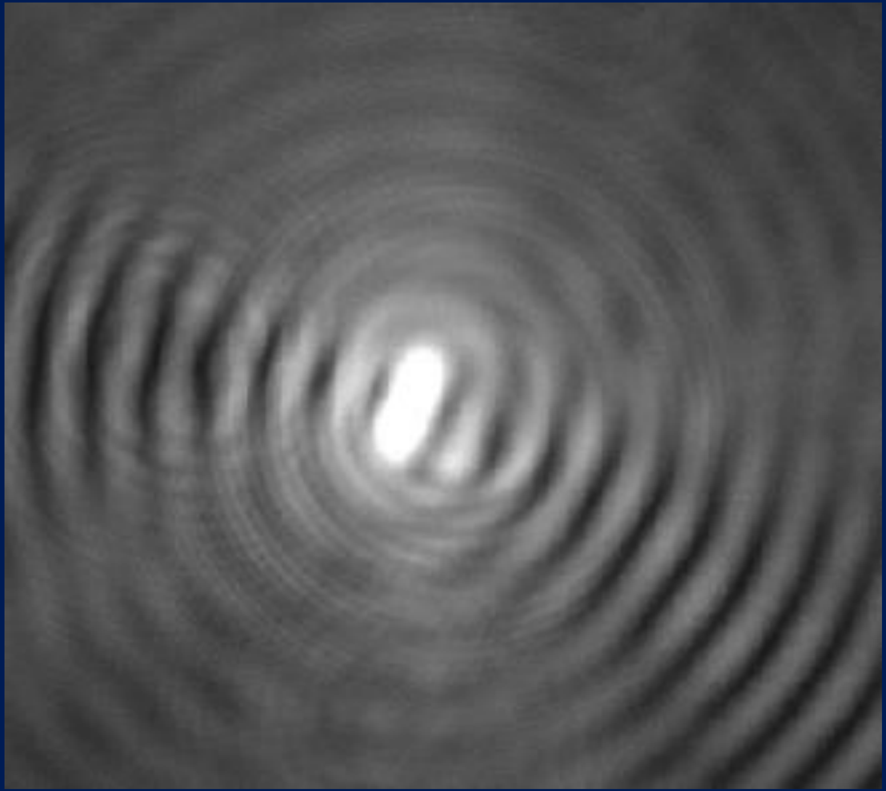
Figure sourced from [2]

Angular to Spatial Chirp at Focus



Angular to Spatial Chirp at Focus (source - Alex)

Lens Focus



Axicon Focus

Why Should You Care?

Angular Chirp

Spatial Chirp at Focus

Pulse Front Tilt

Large Focal Spot Area

Longer Pulse Duration

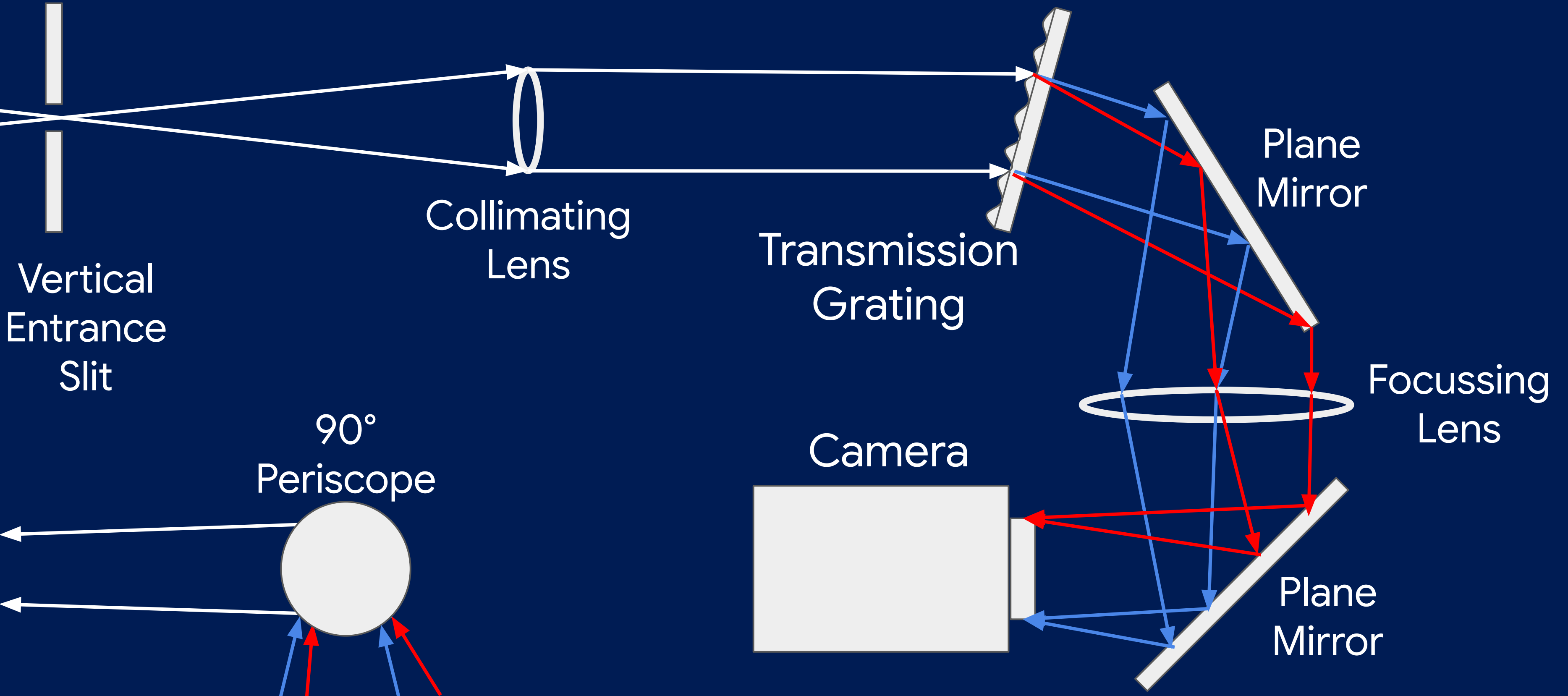
Reduced Intensity

Asymmetric Drive of the Plasma Wave

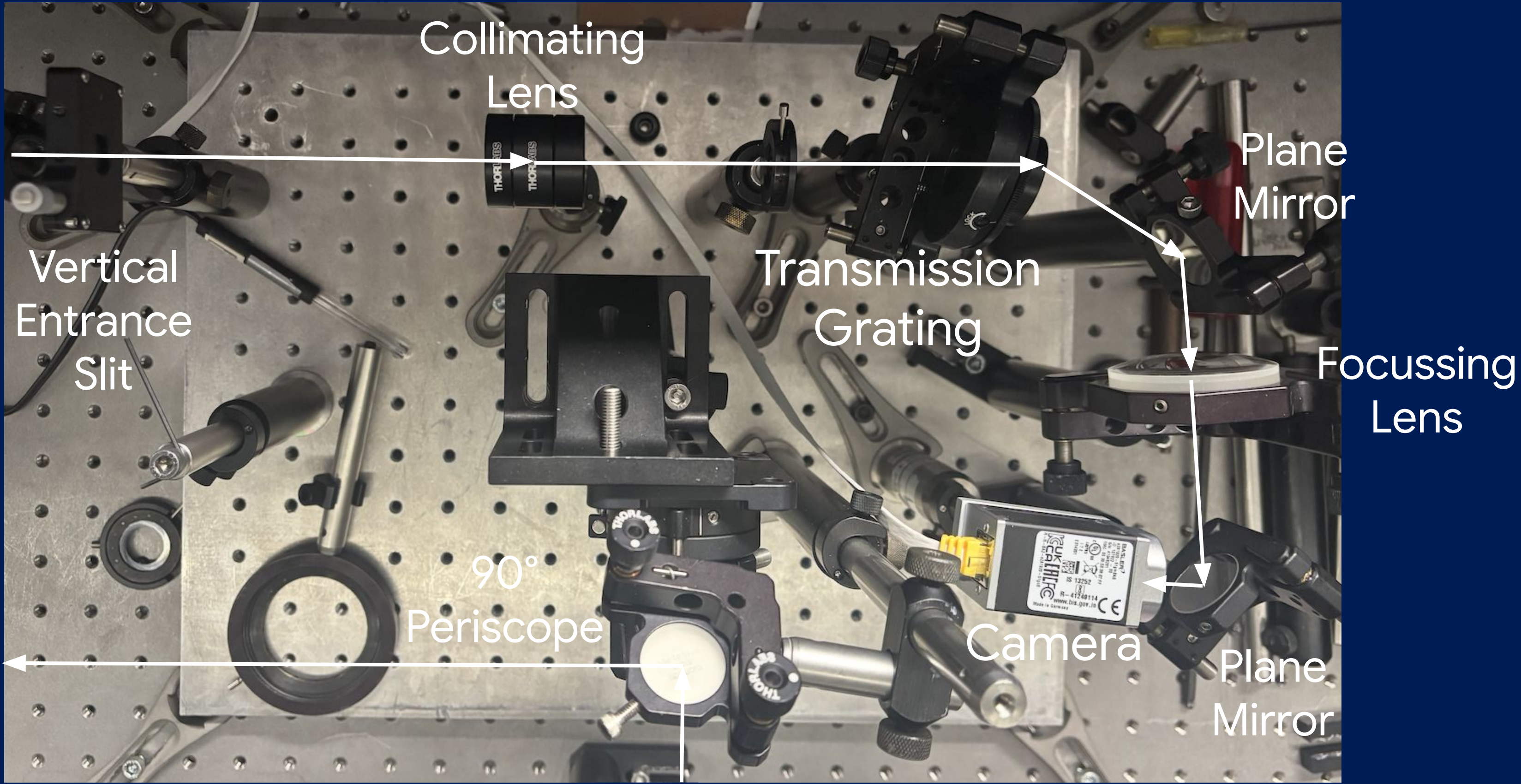
How can we measure angular chirp?

Imaging Spectrometer

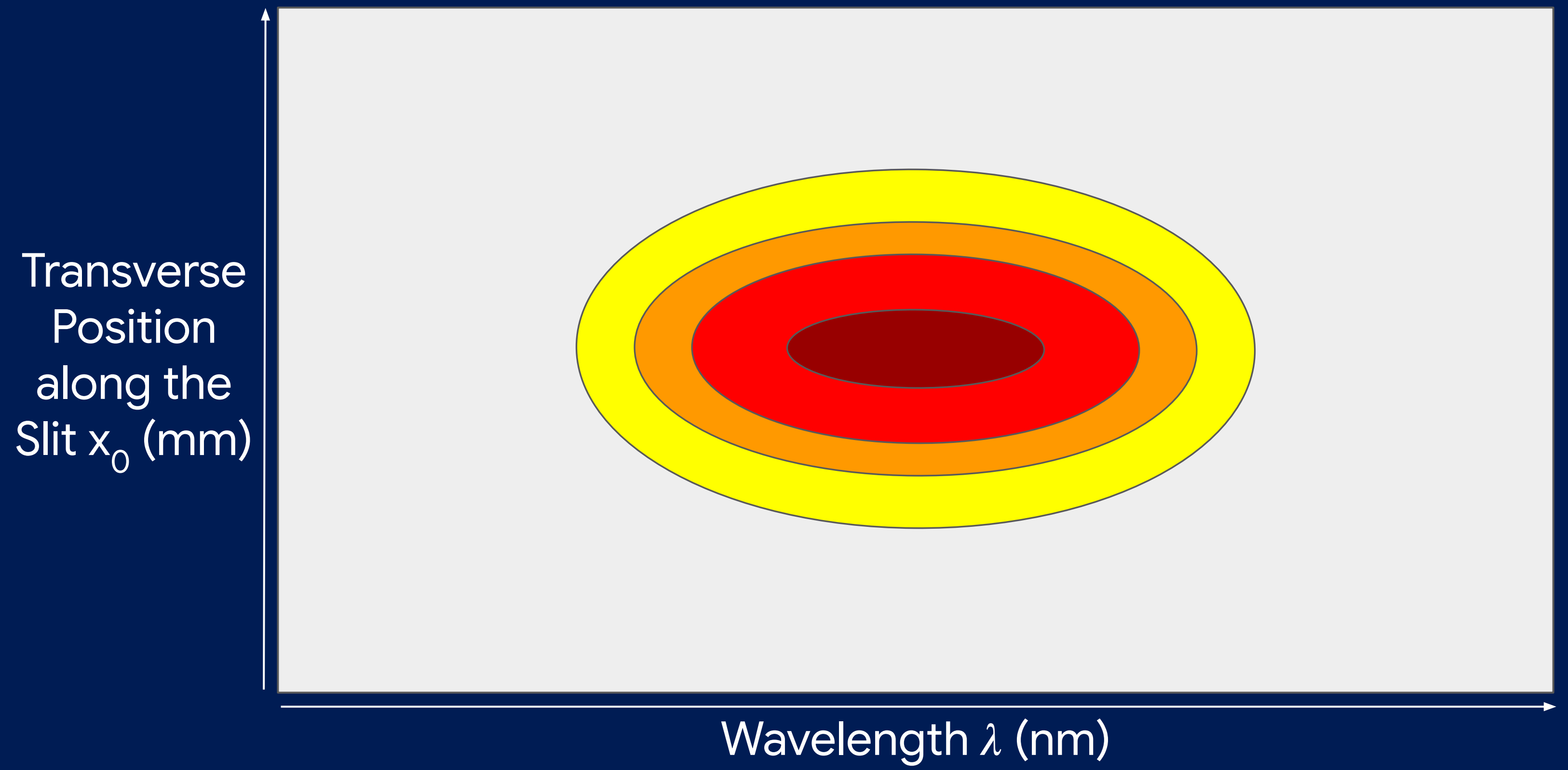
Top View


Vertical
Entrance
Slit
Collimating
Lens
Transmission
Grating
Plane
Mirror
Focussing
Lens
Camera
Plane
Mirror
90°
Periscope

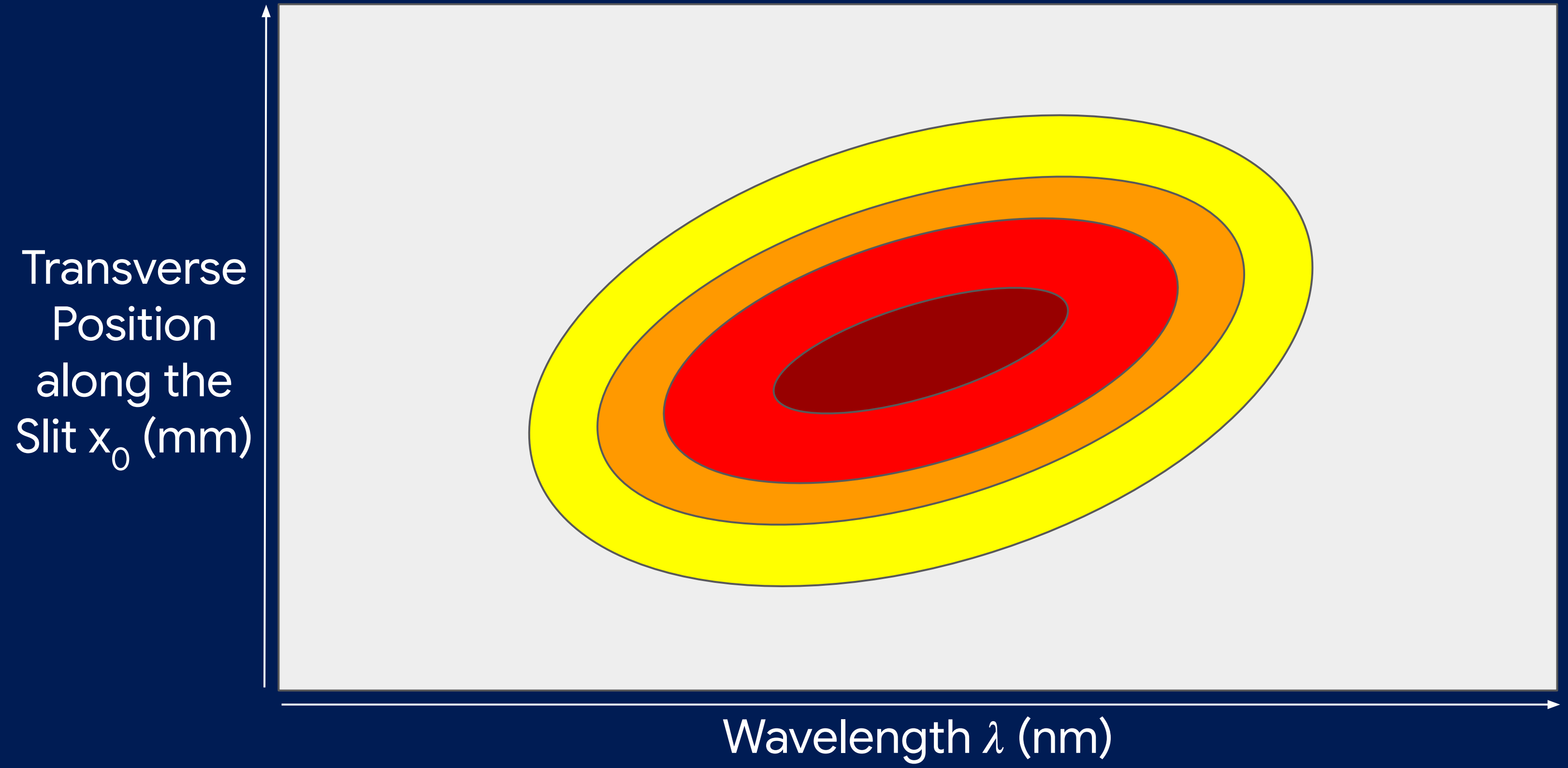
Top View


Collimating Lens
Plane Mirror
Vertical Entrance Slit
Transmission Grating
Focussing Lens
90° Periscope
Camera
Plane Mirror

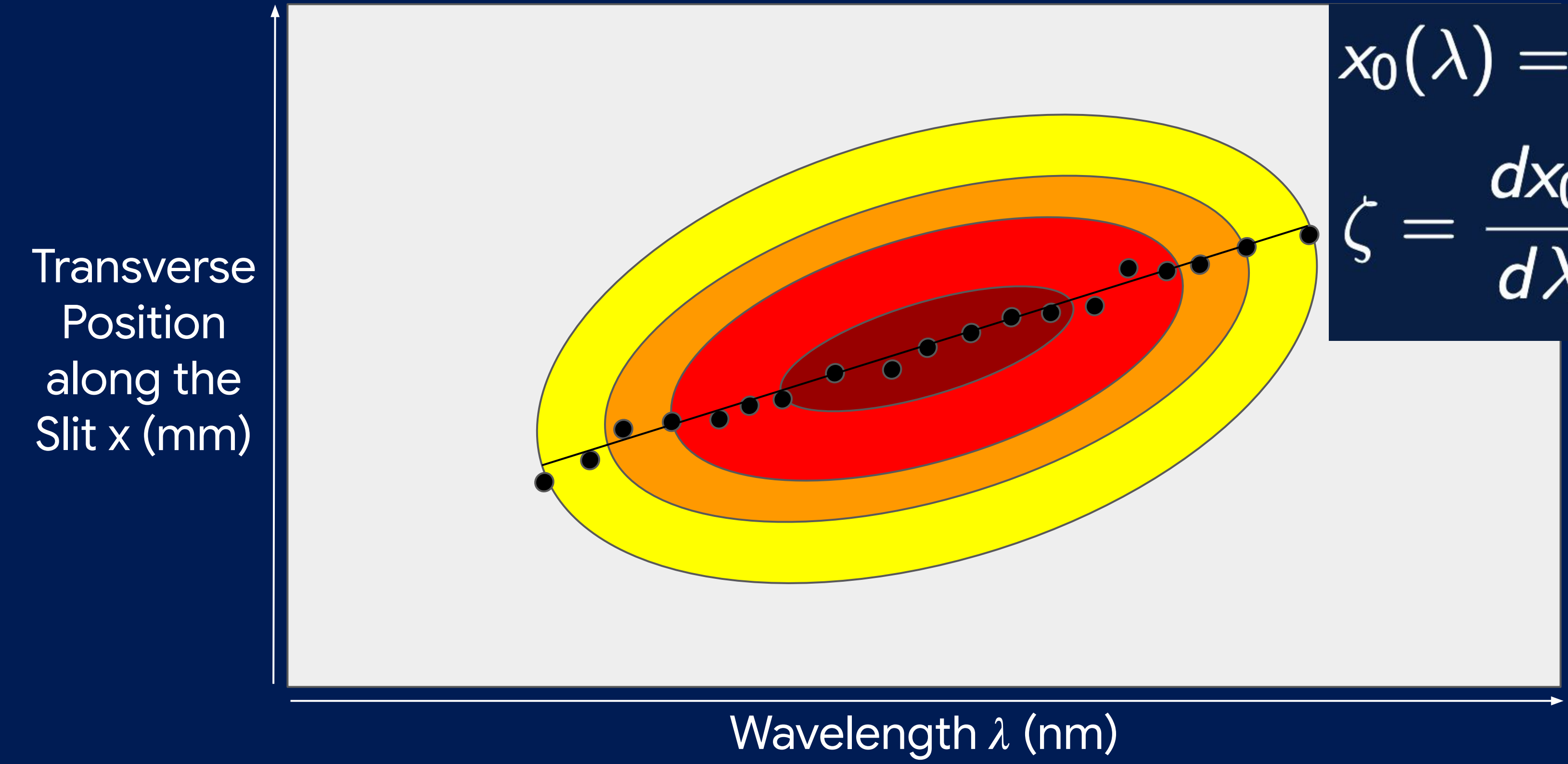
Camera Image (No Spatial Chirp at Entrance Slit)



Camera Image (Spatial Chirp at Entrance Slit)



Camera Image

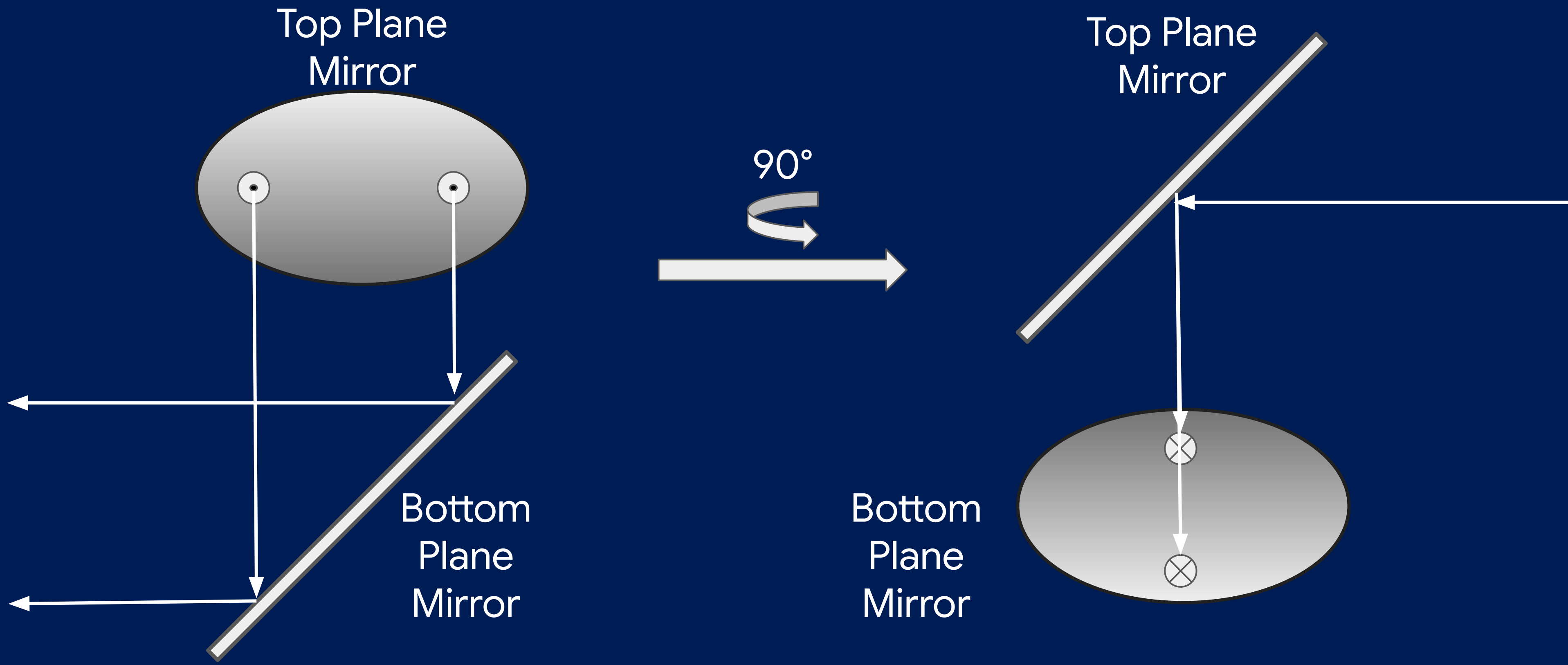


$$x_0(\lambda) = c_0 + c_1\lambda$$

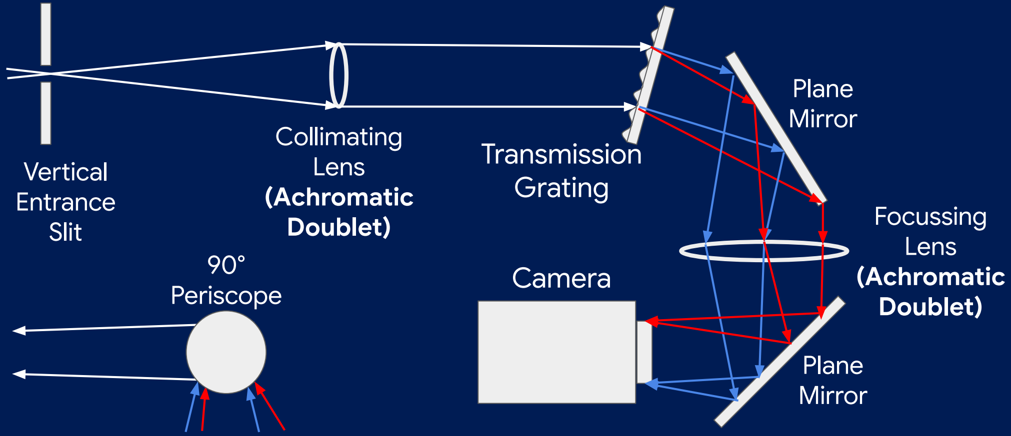
$$\zeta = \frac{dx_0}{d\lambda} = c_1$$

Systematic Considerations

90° Periscope (Side Views)



4f All-Transmissive vs Czerny Turner



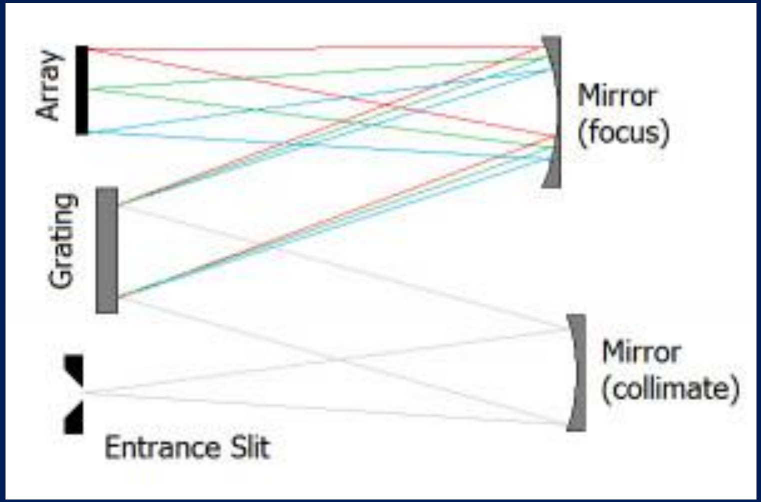
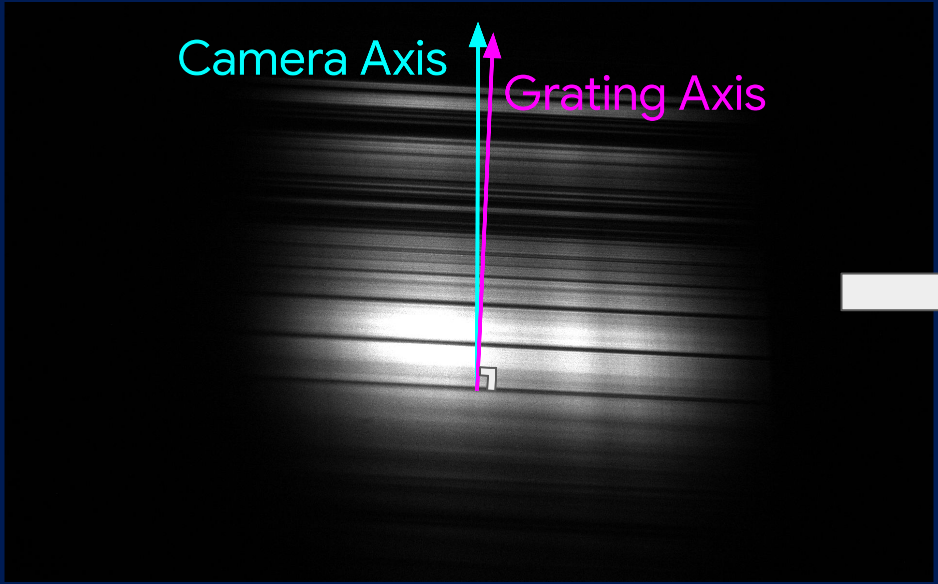


Figure sourced from [3]

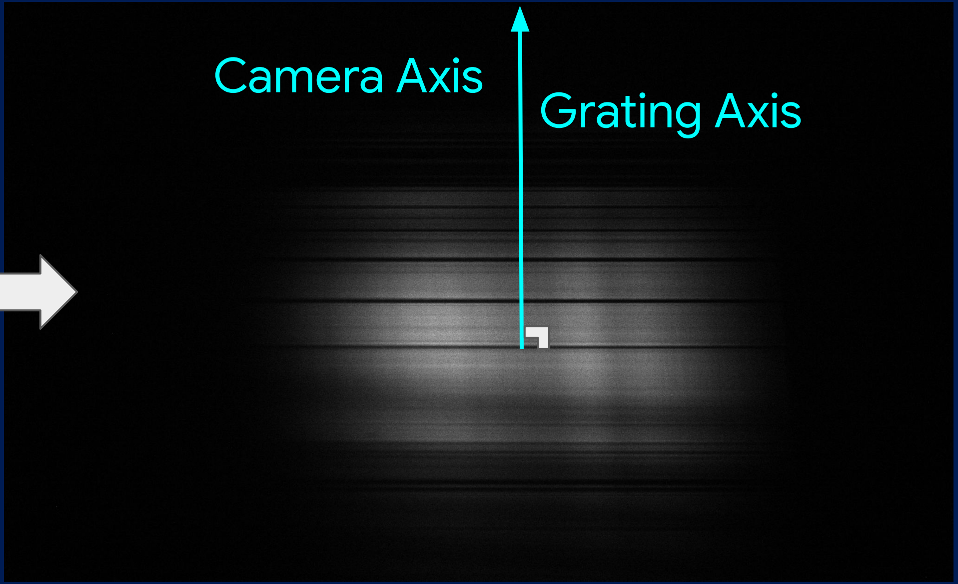
- Simple Geometry
- Less Astigmatism-Prone

- Less Chromatic-Aberration-Prone

Grating-Camera Axis Alignment

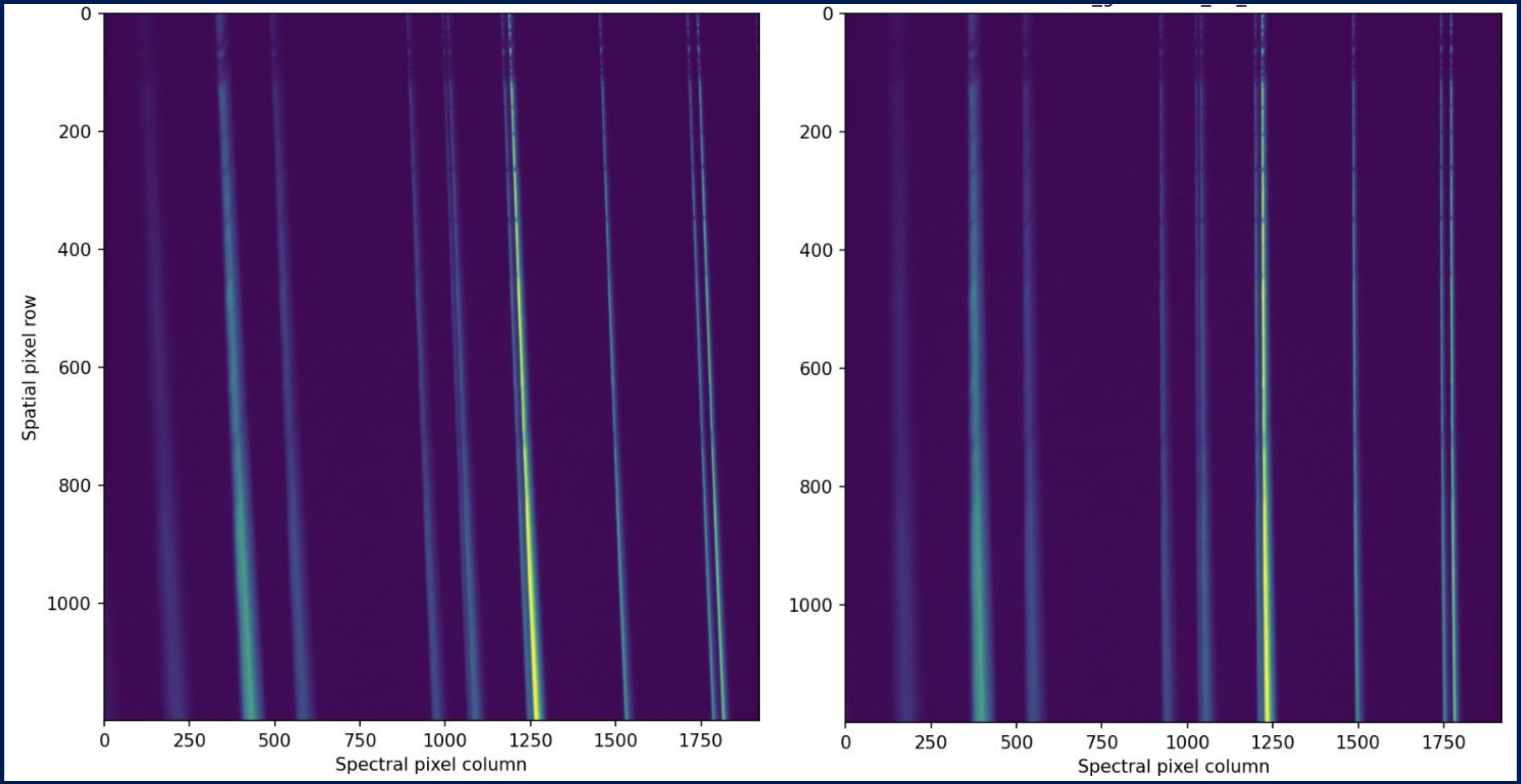


Misaligned

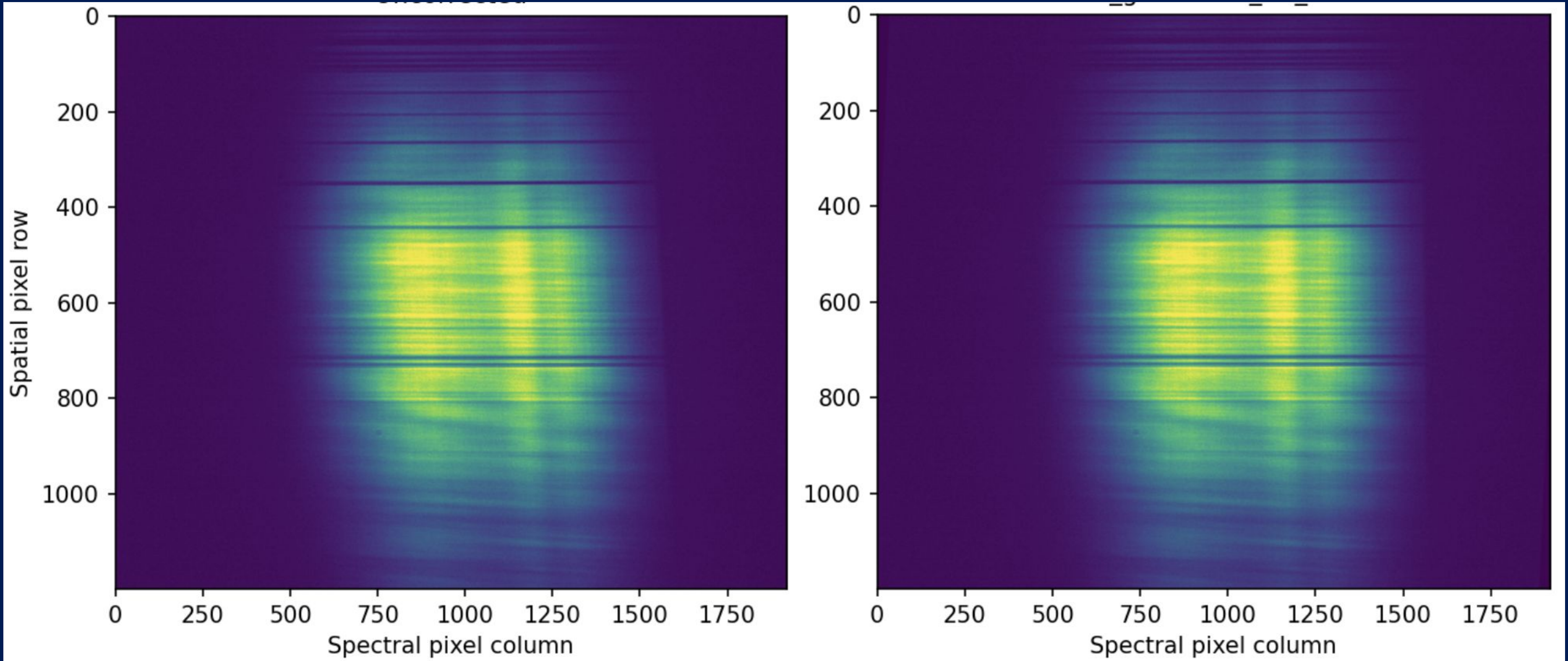


Corrected

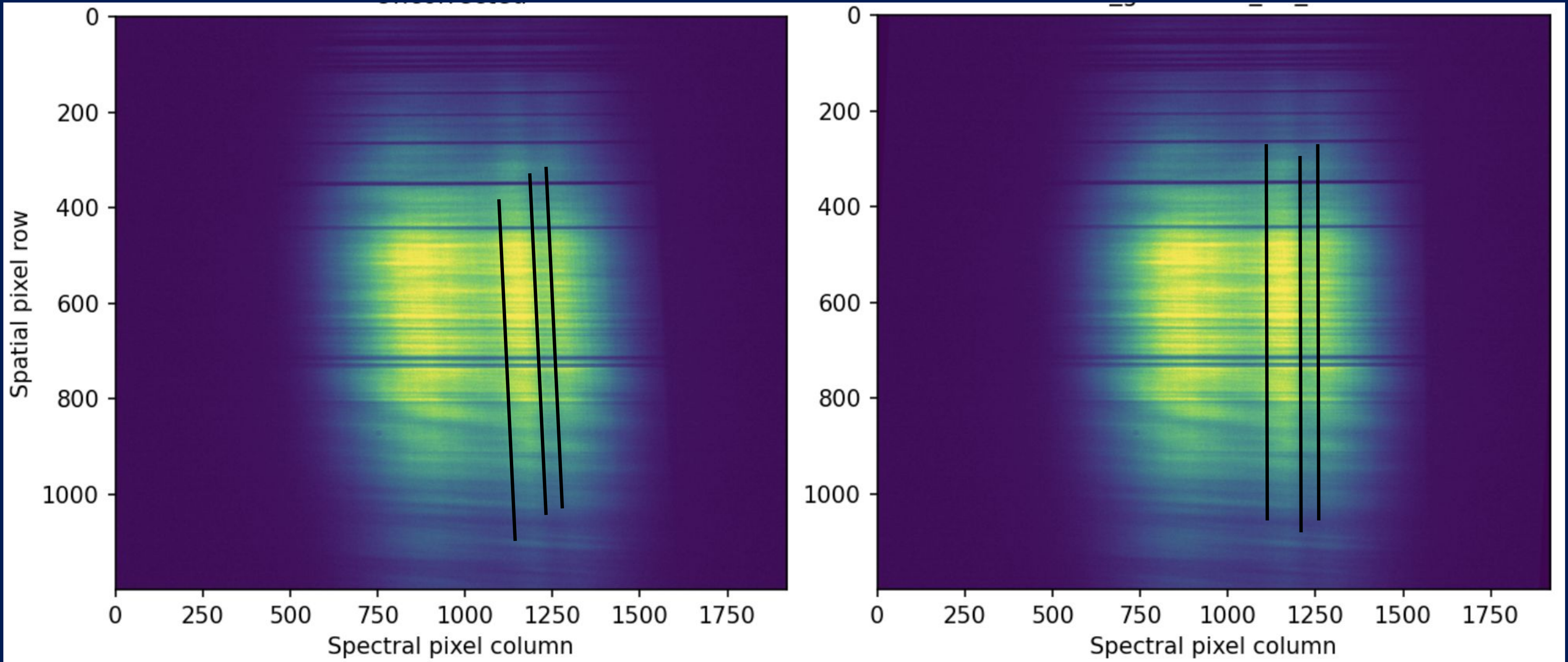
Slit-Camera Axis Alignment



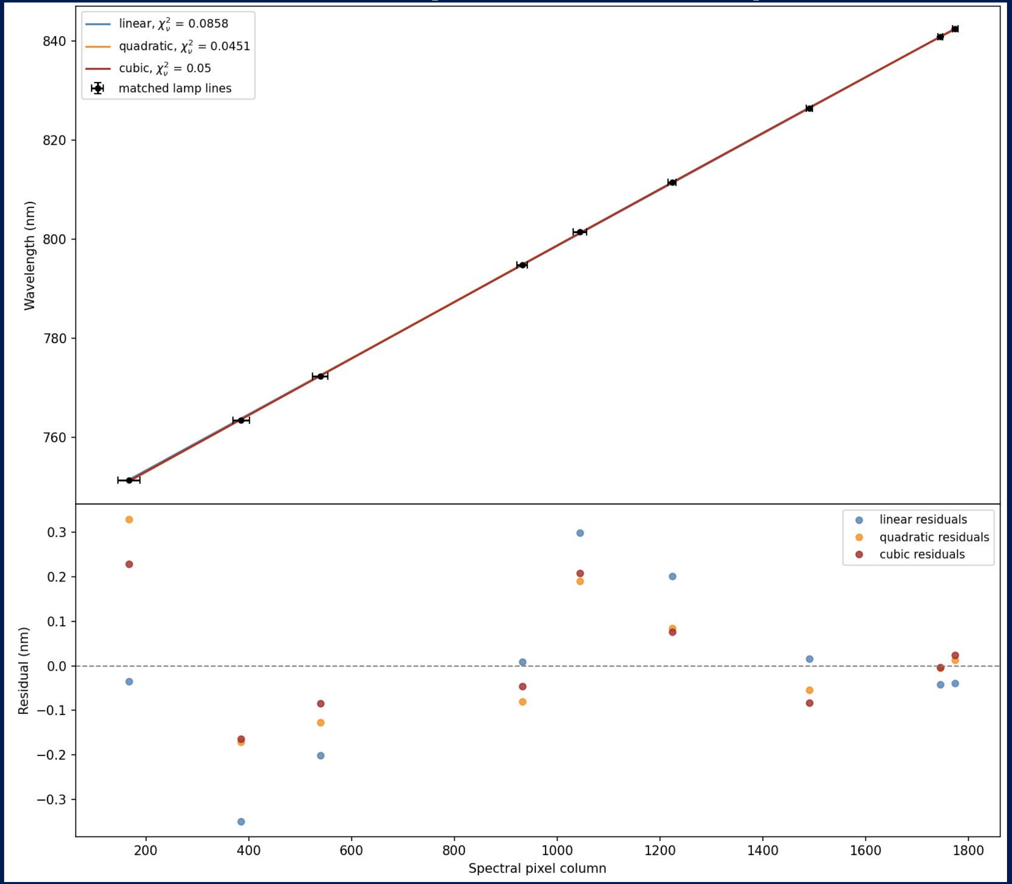
Slit-Camera Axis Alignment



Slit-Camera Axis Alignment

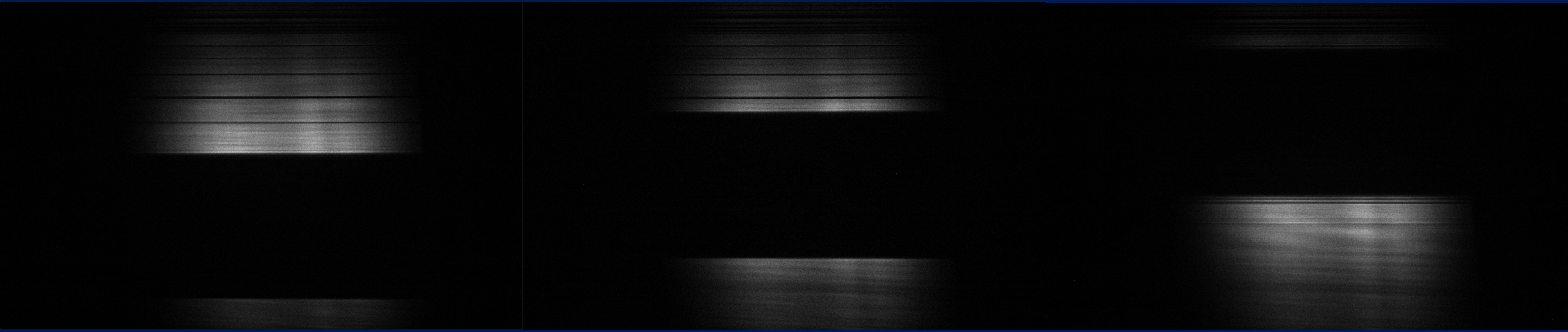


Non-Linear Spectral Dispersion



Note: Spectral pixel column correlated with its uncertainty with r=-0.96

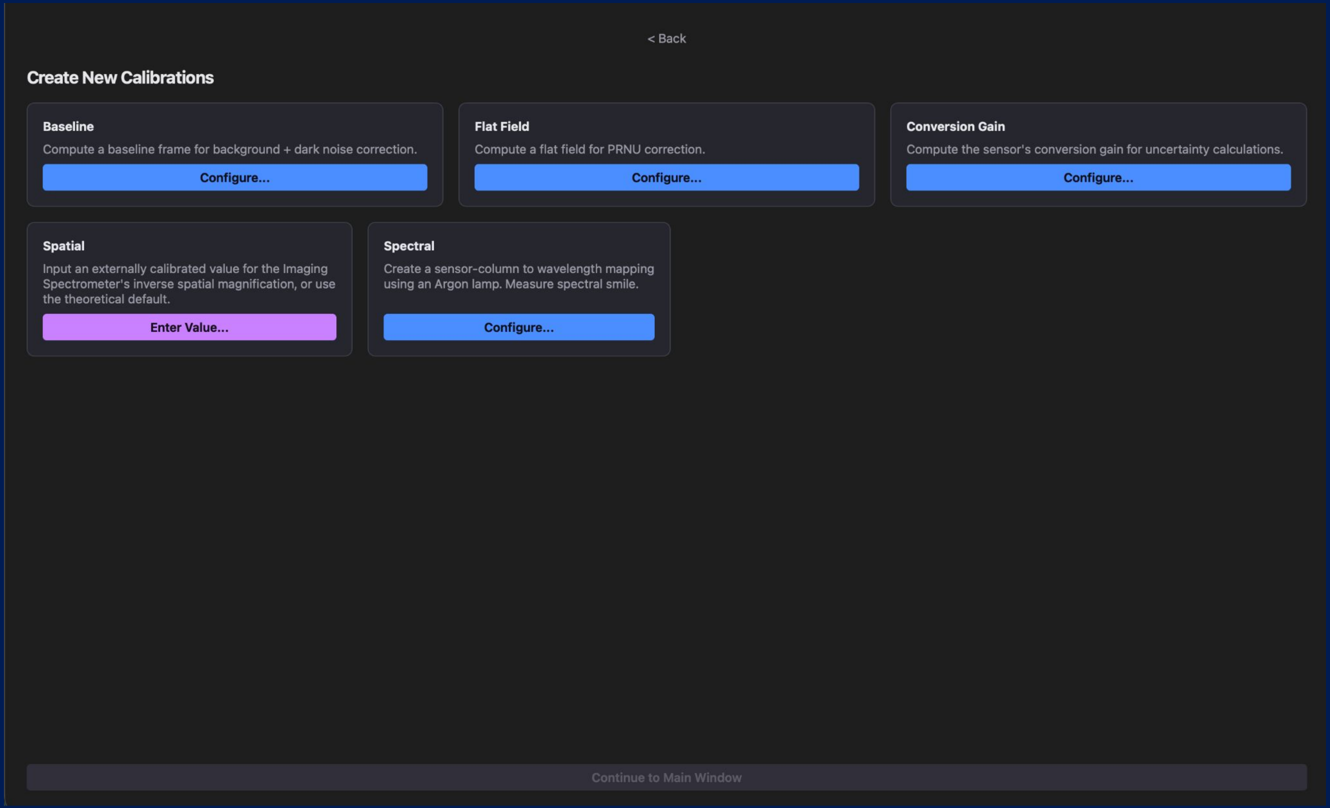
Practical Spatial Axis Magnification



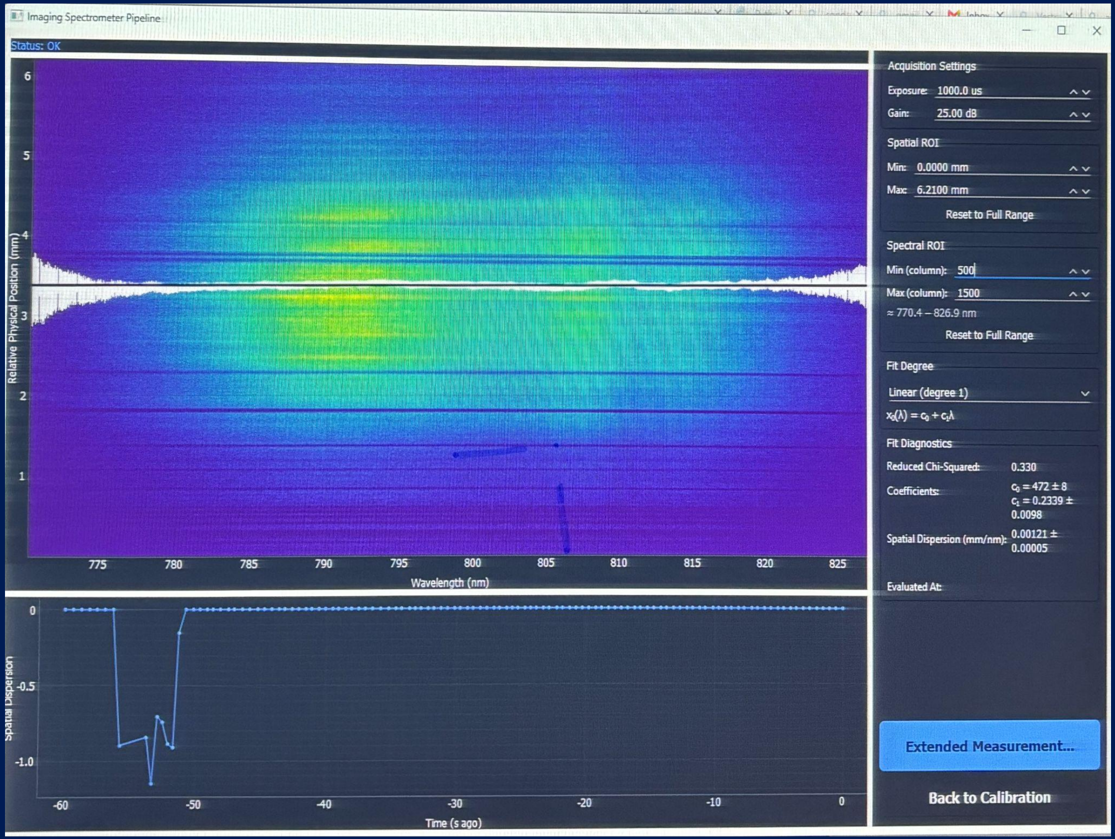
$M = 1.504 \pm 0.008$

Graphical User Interface

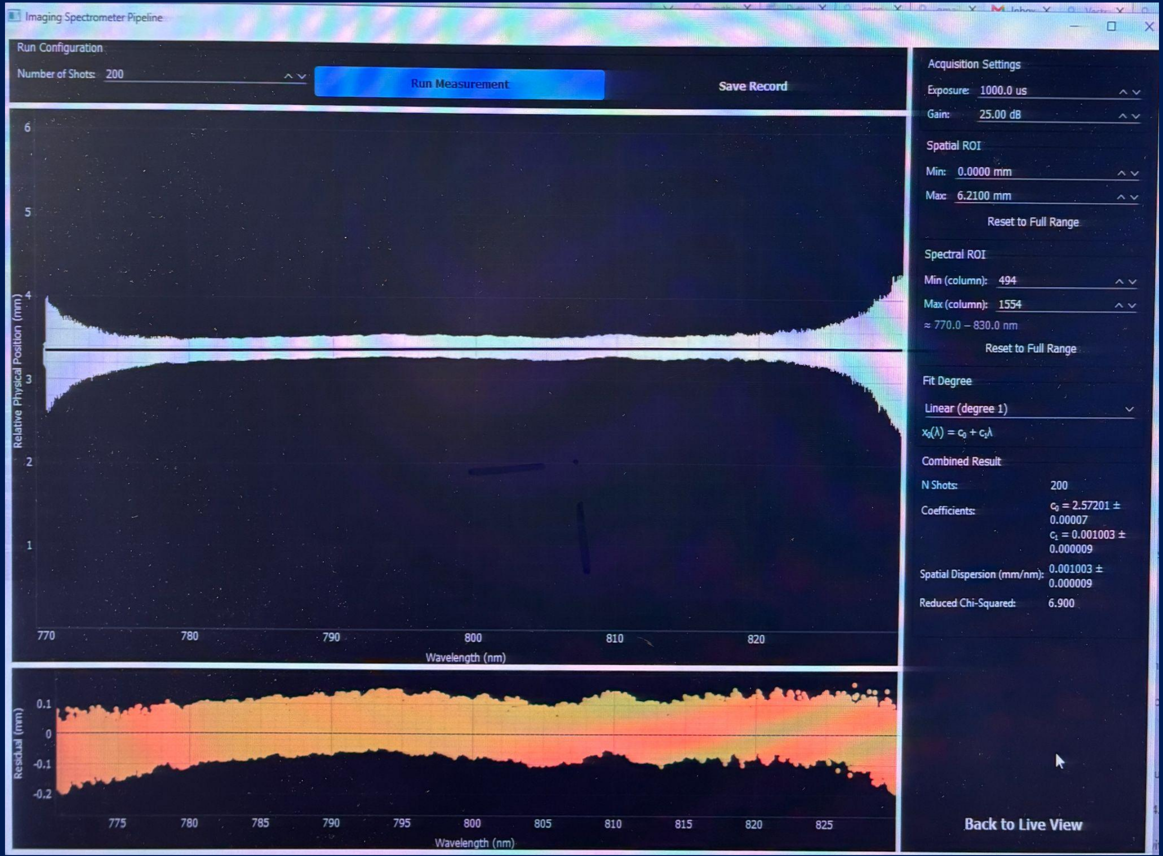
Calibration



Live Measurement

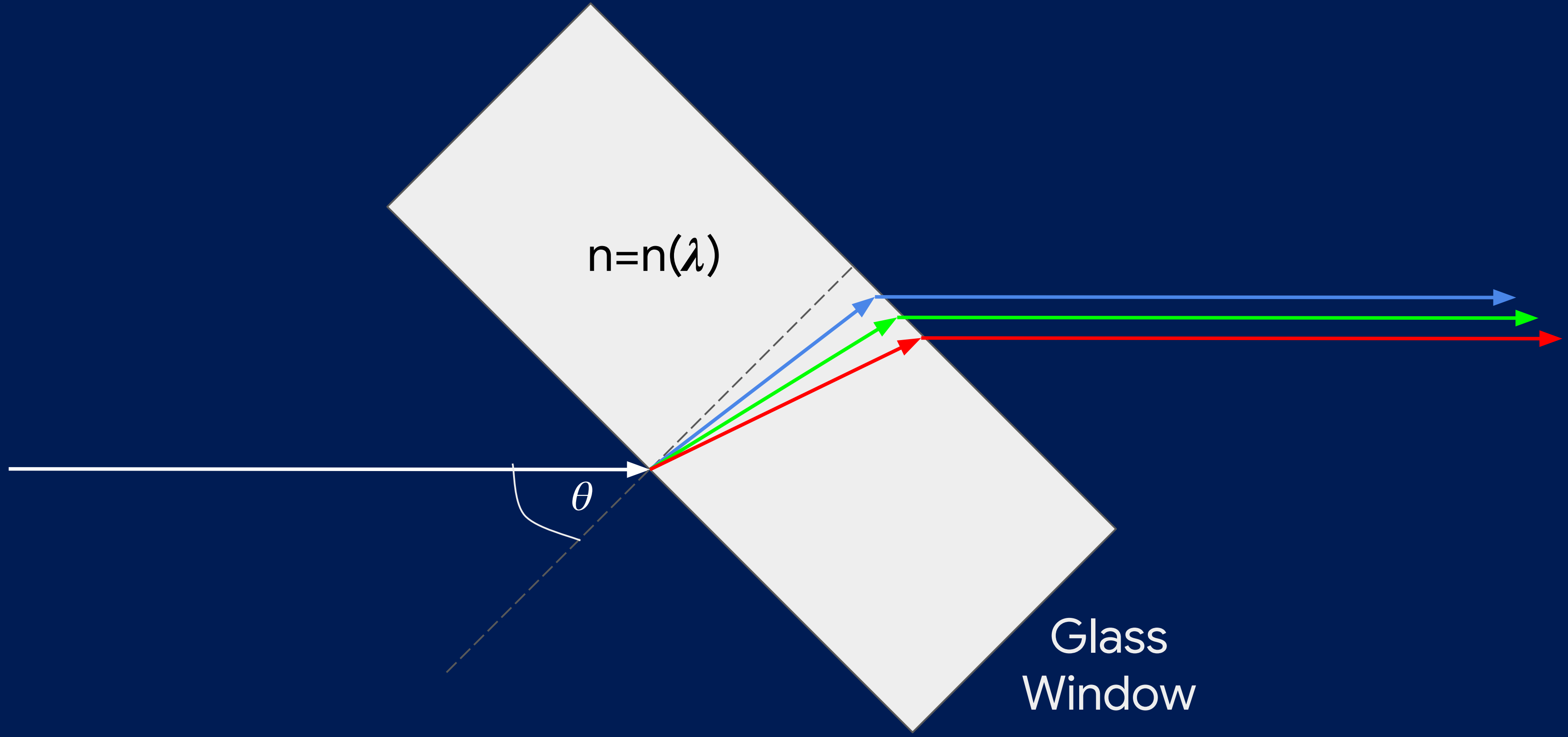


Multi-Shot Measurement



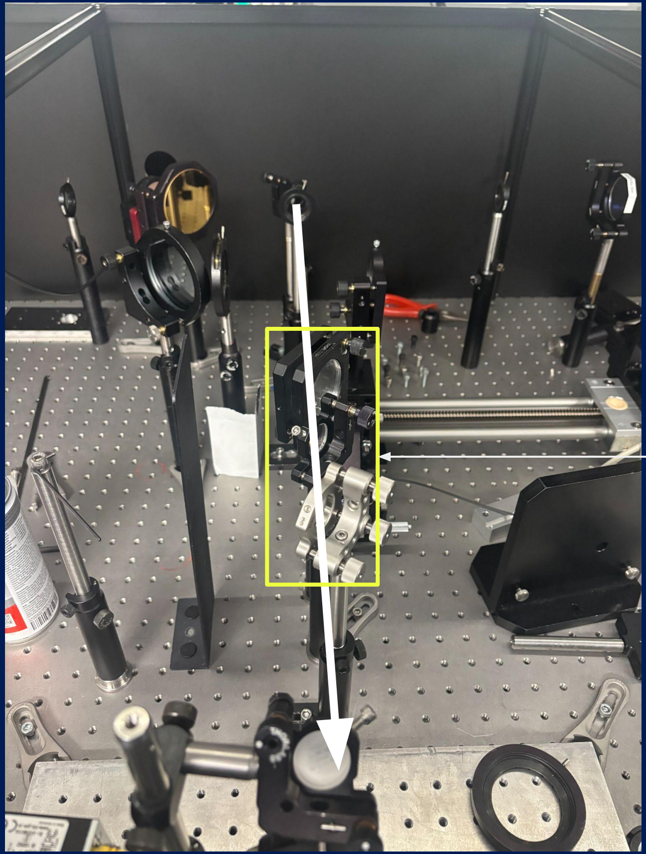
Experiment

Glass-Induced Spatial Chirp



Experimental Design

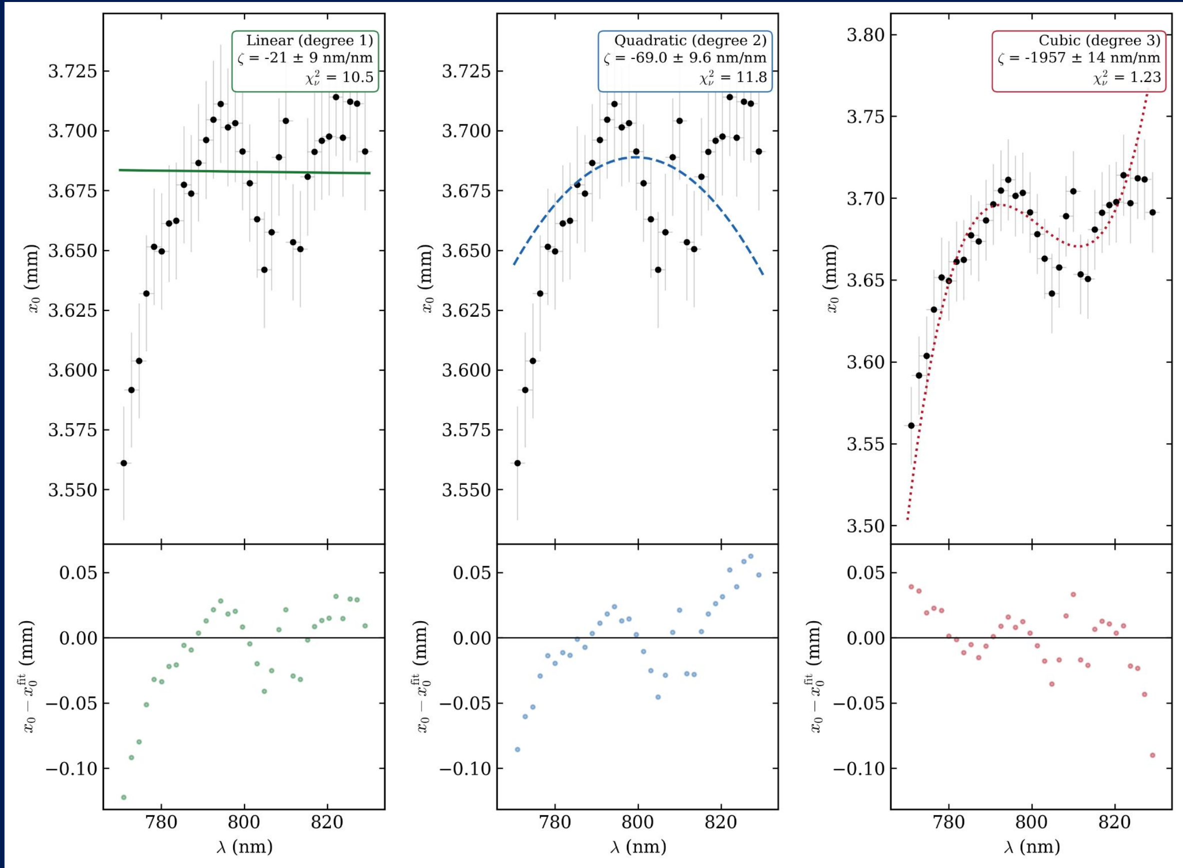
Experimental Design



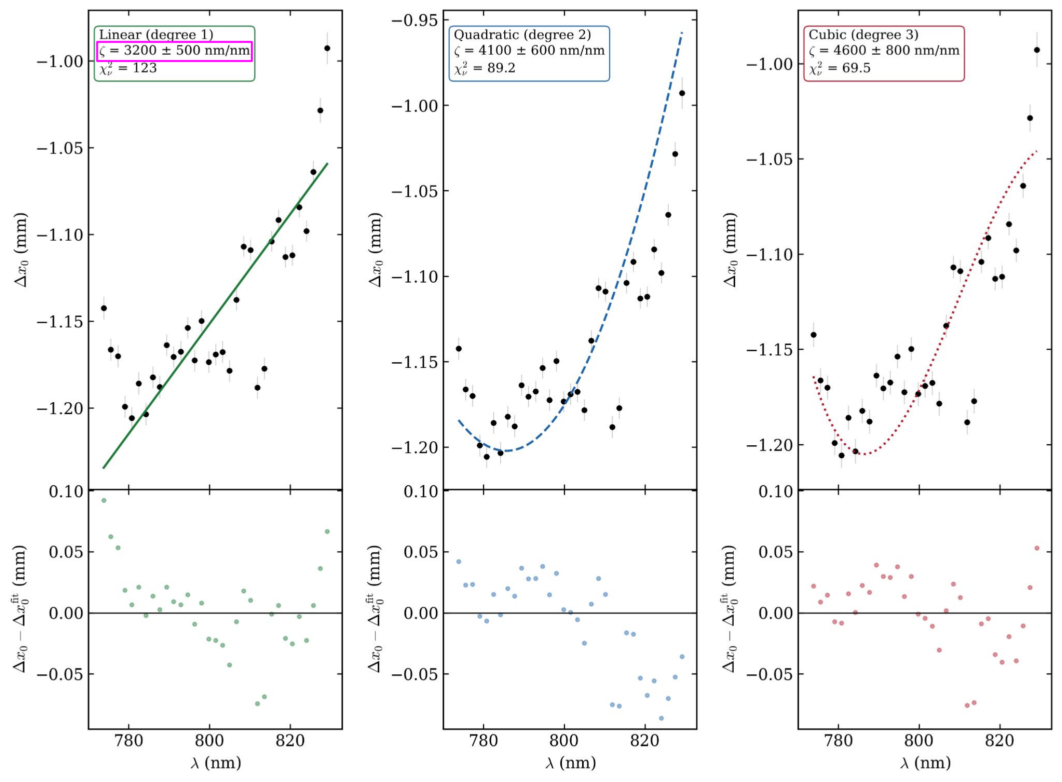
Series of Glass Windows

Results

No Glass Window Baseline

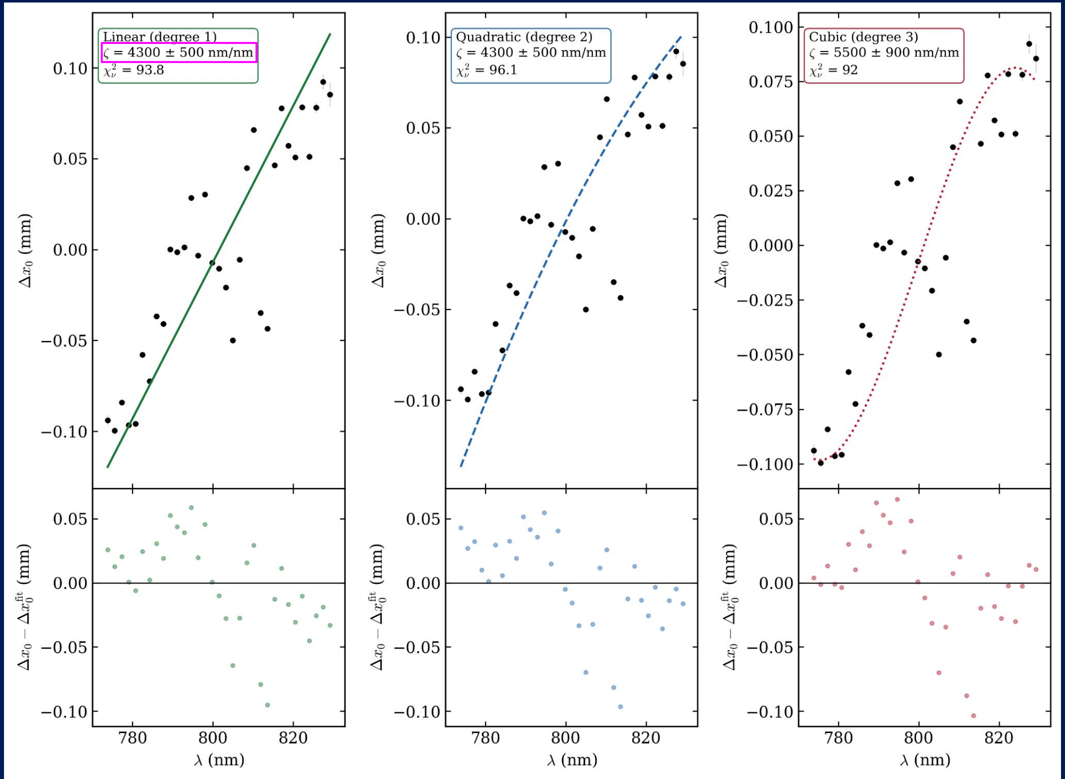


1 Glass Window (Baseline Subtracted) : $\zeta_{\text{predicted}}$ = (81.9 ± 2.8) nm/nm



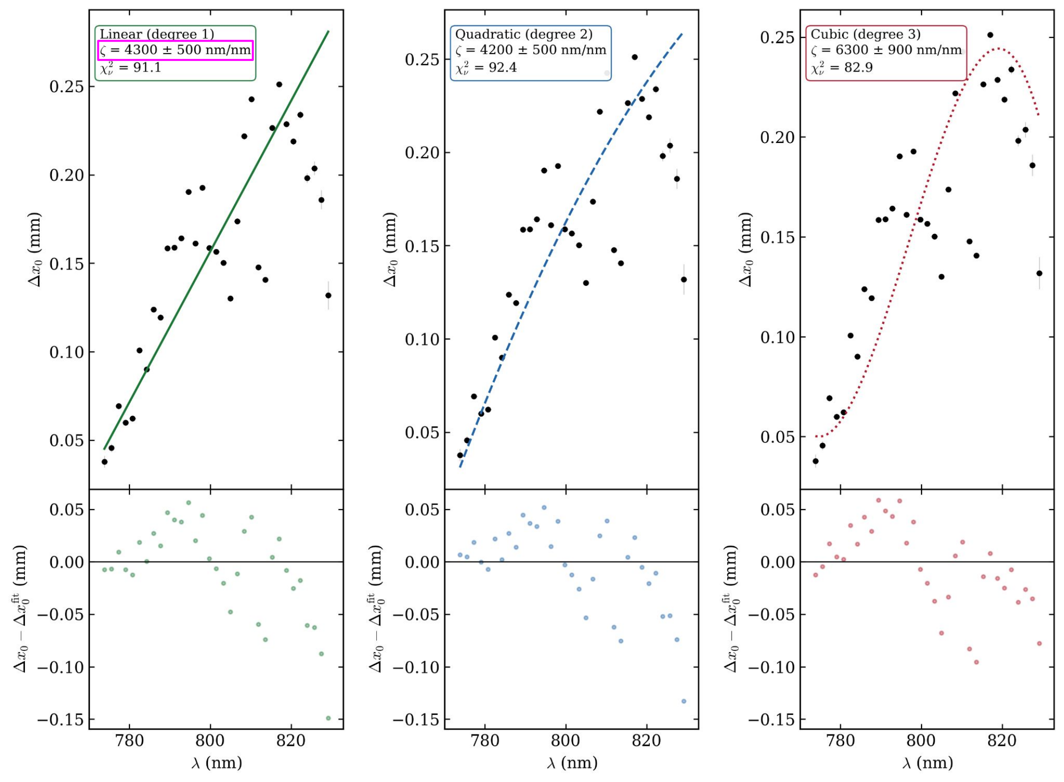
Disagreement of $\approx 6\sigma$

2 Glass Windows (Baseline Subtracted) : $\zeta_{\text{predicted}}$ = (121.3 ± 3.3) nm/nm



Disagreement of ≈ 8σ

3 Glass Windows (Baseline Subtracted) : $\zeta_{\text{predicted}}$ = (134.2 ± 3.4) nm/nm



Disagreement of ≈ 8σ

At Face Value The Order of Magnitudes Imply Either...

Parameter	Implied Value	Real Value
Total Window Thickness	600 mm	20 mm
Refractive Index Difference Over Bandwidth	3×10^{-2}	1×10^{-3}
Refractive Index of Air	1.7	1.0

Discussion

Potential Systematic Errors

Physical

Data Analysis

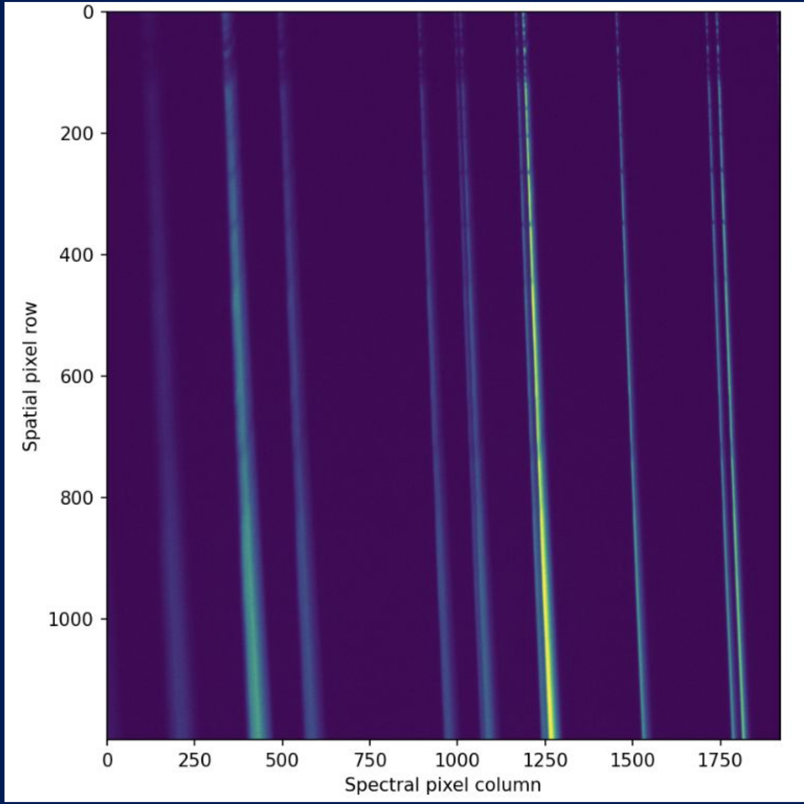
Data Analysis

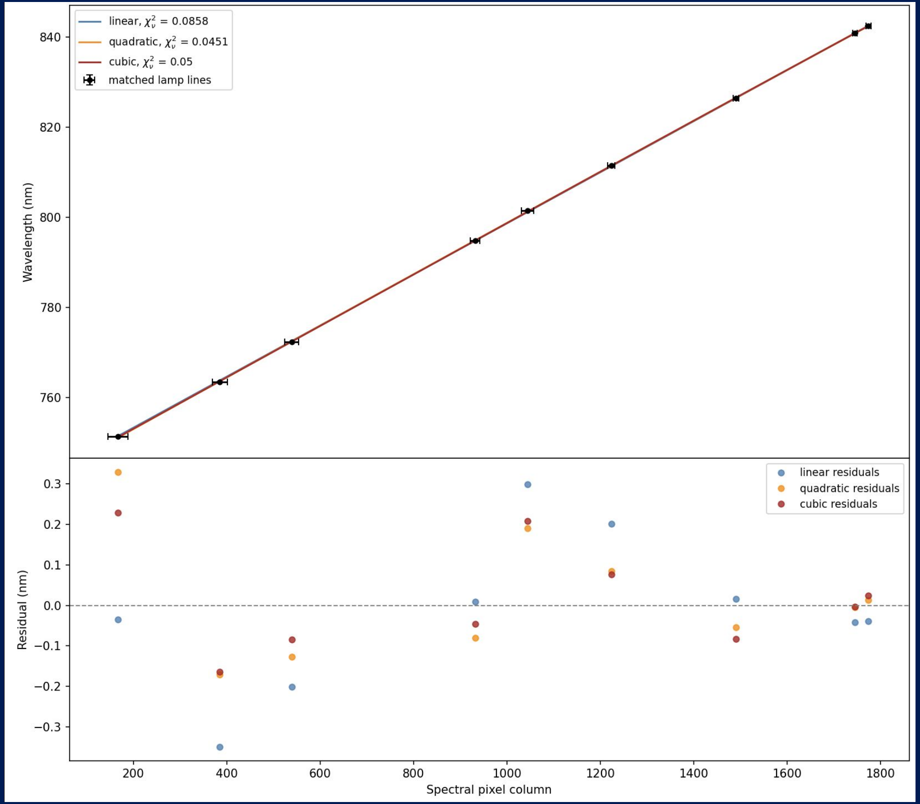
- Finite-difference and fit dispersion directly
- Vary Wavelength bin-widths
- Averaging shots before vs after centroiding

Physical

- Misalignments eg. 90 degree periscope
- Defects
- Chromatic aberration
- Scheimpflug-type tilt
- Residual camera-grating tilt

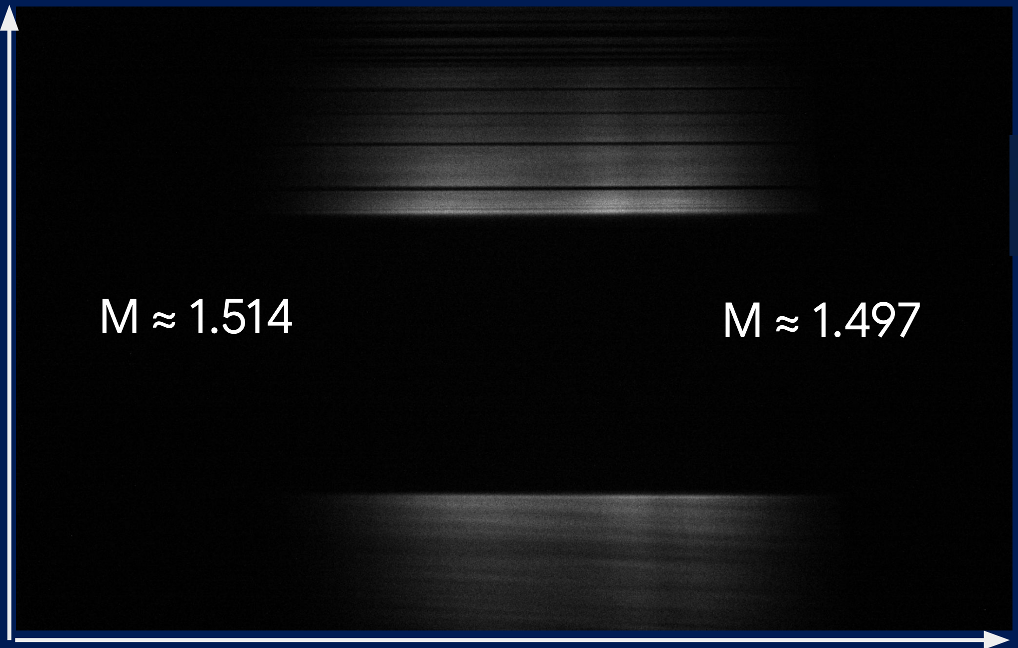
Scheimpflug-type tilt





For an Imperfect 4f System this causes $M = M(\lambda)$

Height along the Slit



Wavelength

$$x_{slit}(\lambda) = M(\lambda)[x_{det}(\lambda) - x_0]$$

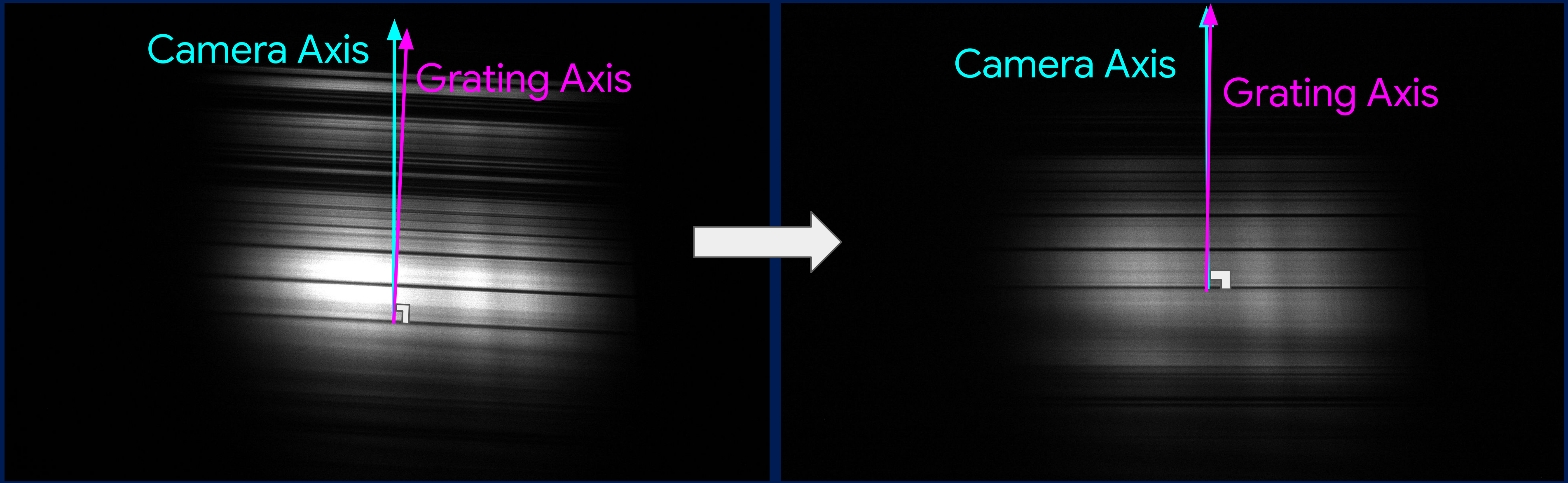
$$\Downarrow$$

$$\frac{dx_{slit}}{d\lambda} = M(\lambda)\frac{dx_{det}}{d\lambda} + (x_{det} - x_0)\frac{dM}{d\lambda}$$

Calls for direct mapping of $x_{det}(\lambda) \longrightarrow x_{slit}(\lambda)$

Need $\Delta M \approx 8$ times larger to explain order of magnitude, dependent on choice of x_0

Residual Camera-Grating Tilt

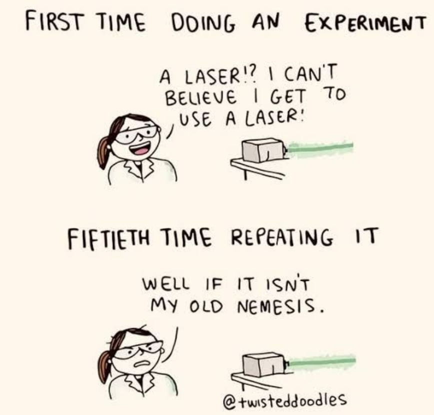


Misaligned

Accounts for
100-400 nm of fake
spatial chirp

Still Misaligned

Conclusions and Future Work



- Spatio-spectral distortions hinder P-MoPA
- Designed and deployed an Imaging Spectrometer
- Attempted to correct systematics
- Built a Graphical User Interface
- Designed an experiment for testing
- Considered and ruled out systematics that may explain result discrepancies
- Future work: Fix systematics, starting with spatial calibration

References

[1] O. Jakobsson, S. M. Hooker, and R. Walczak, "GeV-Scale Accelerators Driven by Plasma-Modulated Pulses from Kilohertz Lasers," Physical Review Letters, vol. 127, no. 18, Art. no. 184801, Nov. 2021, doi: 10.1103/PhysRevLett.127.184801.

[2] Center for Ultrafast Optical Science, University of Michigan, "Chirped Pulse Amplification," Experimental Facilities. Accessed: Aug. 15, 2026. [Online]. Available: https://cuos.engin.umich.edu/researchgroups/hfs/facilities/chirped-pulse-amplification/

[3] Gigahertz-Optik GmbH, "Setup of a Spectroradiometer," Knowledge Base: Basics of Light Measurement. [Online]. Available: https://www.gigahertz-optik.com/en-us/service-and-support/knowledge-base/basics-light-measurement/spectr-radiometers/setup/. Accessed: Aug. 16, 2026.

[4] S. Akturk, X. Gu, P. Bowlan, and R. Trebino, “Spatio-temporal couplings in ultrashort laser pulses,” J. Opt., vol. 12, no. 9, Art. no. 093001, 2010.